



BAAZ JKBOSE

🔥 2026 CLASS 11th 🔥

SETS

MATHS

LECTURE-02

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TOPICS TO BE COVERED

1

Representation of Sets

$\in \rightarrow$ Belongs To
(epsilon)

$\notin \rightarrow$ Don't belongs To

2

Exercise Discussion 1.1

* $a \in A \Rightarrow a$ is an element of A

* $a \notin A \Rightarrow a$ is not an element of A .

Quick Recap:

- * Well defined collection of objects.
- * Members of the set are known as the elements of set.
- * The no. of elements in a set is known as the Cardinality of the set. $\rightarrow n(A) / |A|$

$|A|, n(A) \rightarrow$ Cardinality of set A

Representation of a Set

There are two ways for representing a set:

* Roster form / Tabular form:

In this form, we list all the elements of the set, separate them by commas & are enclosed within Brackets $\{ \}$

- A : Set of Primes b/w 1 & 30.

$$\Rightarrow A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$$

- B : Set of vowels in English alphabets

$$\Rightarrow B = \{a, e, i, o, u\}$$

- C : Set of Natural numbers b/w less than 10.

$$\Rightarrow C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

* Set-Builder form/Property Method:

In this form, we represent all the elements of the set by means of variable, then followed by the property satisfied by each element of the set.

• $A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

$\Rightarrow A = \{x : x \text{ is a prime b/w } 1 \text{ \& } 30\}$

Tawkeer $\rightarrow \{T, A, K, E, R\}$

• $B = \{2, 4, 6, 8, 10, 12\}$

$2 \times 1 \quad 2 \times 2 \quad 2 \times 3 \quad 2 \times 4 \quad 2 \times 5 \quad 2 \times 6$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$

$\Rightarrow B = \{x : x = 2n, n \in \mathbb{N} \text{ \& } n \leq 6\}$

• $C = \{2, 4, 8, 16, 32, 64\}$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2^1 \quad 2^2 \quad 2^3 \quad 2^4 \quad 2^5 \quad 2^6$

$\Rightarrow C = \{x : x = 2^n, n \in \mathbb{N} \text{ \& } n \leq 6\}$

Example

Write down the following sets in the set-Builder form:

(a) $\{3, 6, 9, 12\}$

(b) $\{2, 4, 8, 16, 32\} \xrightarrow{\text{Set Builder form}} \{x : x = 2^n, n \in \mathbb{N} \text{ \& } n \leq 5\}$

(c) $\{1, 4, 9, 16, \dots, 100\} \xrightarrow{\text{Set Builder form}} \{x : x = n^2, n \in \mathbb{N} \text{ \& } n \leq 10\}$

(Sol) (a) $\{3, 6, 9, 12\} \xrightarrow{\text{Set Builder form}} \{x : x = 3n, n \in \mathbb{N} \text{ \& } n \leq 4\}$

$3 \times 1 \quad 3 \times 2 \quad 3 \times 3 \quad 3 \times 4$

$* \{1^2, 4, 9, 16, 25, \dots, 100\}$

Question

The Set of all Letters of the Word **"MATHEMATICS"** in roster form is :

~~1~~ { M, T, A, H, M, I, C, S } ✗

~~2~~ { M, A, T, H, E, T, I, C, S } ✗

~~3~~ { A, M, T, E, H, I, C, S } ✓

~~4~~ { MA TH, E, I, C, S } ✗

QuestionRoster form

The Set $A = \{x : x \text{ is Positive prime} < 10\}$ in the tabular form is

☒ 1 $\{1, 2, 3, 5, 7\}$ ✗

☒ 2 $\{1, 3, 5, 7, 9\}$ ✗

☒ 3 $\{2, 3, 5, 7\}$ ✓✓✓✓✓

☒ 4 None of these

Question

The Set $A = \{ 1, 8, 27, 64, 125 \}$ in the set Builder form is

$$\begin{array}{ccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1^3 & 2^3 & 3^3 & 4^3 & 5^3 \end{array}$$

~~1~~ $\{n^2 : n \in \mathbb{N} \text{ and } n \leq 6\}$. ~~X~~

~~2~~ $\{n^3 : n \in \mathbb{N} \text{ and } n \leq 5\}$.

~~3~~ $\{n^4 : n \in \mathbb{N} \text{ and } n \leq 6\}$.

~~4~~ $\{2n : n \in \mathbb{N} \text{ and } n \leq 6\}$.

$$\{x : x = n^3, n \in \mathbb{N} \text{ and } n \leq 5\}$$

Question

Which of the following is a set Builder form of the set $\{1, 3, 5, 7, 9, 11, 13, 15\}$?

~~1~~

$\{x : x = 2n + 1, n \in \mathbb{N} \text{ and } n \leq 8\}$. ~~X~~

$2 \times 8 + 1 = 17$

~~2~~

$\{x : x = 2n + 1, n \in \mathbb{N} \text{ and } n \leq 7\}$. ~~X~~

~~3~~

$\{x : x = 3n + 1, n \in \mathbb{N} \text{ and } n \leq 8\}$. ~~X~~

~~4~~

$\{x : x = \underline{2n + 1}, n \in \mathbb{W} \text{ and } n \leq 7\}$.

$2 \times 0 + 1$ $2 \times 1 + 1$

$1 = 2 \times 0 + 1$

$3 = 2 \times 1 + 1$

$5 = 2 \times 2 + 1$

$7 = 2 \times 3 + 1$

$9 = 2 \times 4 + 1$

$11 = 2 \times 5 + 1$

$13 = 2 \times 6 + 1$

$15 = 2 \times 7 + 1$

odd numbers ($2n + 1$)

Question

Which of the following is a set Builder form of the set $\{-1, 0, 1, 2, 3, 4, 5, 6, 7\}$?

$$\{x : x \in \mathbb{Z}, -2 < x < 8\}$$

~~1~~ $\{x : -1 < x < 7, x \in \mathbb{R}\}$ ~~X~~

~~2~~ $\{x : -1 \leq x < 7, x \in \mathbb{N}\}$ ~~X~~

~~3~~ $\{x : -2 < x < 8, x \in \mathbb{Z}\}$ ✓

~~4~~ $\{x : -1 \leq x \leq 7, x \in \mathbb{R}\}$ ~~X~~

Question

Which of the following is a set Builder form of the set $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right\}$? $\left\{\frac{n}{n+1}, n \in \mathbb{N} \text{ and } n \leq 3\right\}$

- 1 $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } n > 1\right\}$. ✗
- 2 $\left\{x : x = \frac{n-1}{n+1}, n \in \mathbb{N} \text{ and } n > 2\right\}$. ✗
- 3 $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 < n < 2\right\}$. ✗
- 4 $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N} \text{ and } 1 \leq n < 4\right\}$. ✓

Question

If $A = \{x : x^2 - x - 6 = 0 \text{ and } x \in \mathbb{N}\}$, then the set A in Roster form is

~~1~~ $\{3\}$ ✓ ✓ $= \{3\}$

~~2~~ $\{2\}$

3 $\{3, -2\}$ ✗

4 $\{1, 2\}$ ✗

$$x^2 - x - 6 = 0$$

$$\Rightarrow x^2 - 3x + 2x - 6 = 0$$

$$\Rightarrow x(x - 3) + 2(x - 3) = 0$$

$$\Rightarrow (x + 2)(x - 3) = 0$$

$$\Rightarrow x = -2, 3$$

Question

Which of the following statement is false.

$$21 \rightarrow 1, 3, 7, 21.$$

$$28$$

$$6 \rightarrow 1, 2, 3, 6$$

$$1+2+3+6=12=2 \times 6.$$

$$28 \rightarrow 1, 2, 4, 7, 14, 28$$

$$1+2+4+7+14+28=56=2 \times 28.$$

~~1~~ $21 \in \{x : x \text{ has exactly four positive factors}\}$ ✗

~~2~~ $28 \in \{y : \text{the sum of all the positive factors of } y \text{ is } 2y\}$ ✗
Perfect numbers.

~~3~~ $3, 7 \in \{x : x \text{ satisfies the equation } x^2 - 10x + 21 = 0\}$ ✗
 $(x-3)(x-7)=0 \Rightarrow x=3, 7$

~~4~~ $2, 5 \in \{x : x \text{ satisfies the equation } x^2 - 5x + 6 = 0\}$ ✓
 $(x-2)(x-3)=0$
 $\Rightarrow x=2, 3.$

EXERCISE – 1.1

Question

Which of the following are sets? Justify your answer.

- (i) The collection of all the months of a year beginning with the letter J. ✓
- (ii) The collection of ten most talented writers of India. ✗
- (iii) A team of eleven best-cricket batsmen of the world. ✗
- (iv) The collection of all boys in your class. ✓
- (v) The collection of all natural numbers less than 100. ✓
- (vi) A collection of novels written by the writer Munshi Prem Chand. ✓
- (vii) The collection of all even integers. ✓
- (viii) The collection of questions in this Chapter. ✓
- (ix) A collection of most dangerous animals of the world ✗

Question

Let $A = \{1, 2, 3, 4, 5, 6\}$. Insert the appropriate symbol $\in$ or $\notin$ in the blank spaces:

(i) $5 \in A$

(ii) $8 \notin A$

(iii) $0 \notin A$

(iv) $4 \in A$

(v) $2 \in A$

(vi) $10 \notin A$

Question

Write the following sets in **roster form**:

- (i) $A = \{x : x \text{ is an integer and } -3 \leq x < 7\} = \{-3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$.
- (ii) $B = \{x : x \text{ is a natural number less than 6}\} = \{1, 2, 3, 4, 5\}$
- (iii) $C = \{x : x \text{ is a two-digit natural number such that the sum of its digits is 8}\}$
- (iv) $D = \{x : x \text{ is a prime number which is divisor of 60}\}$ H/w $\rightarrow \{17, 71, 26, 62, 35, 53, 44, 810\}$
- (v) $E = \text{The set of all letters in the word TRIGONOMETRY}$ H/w.
- (vi) $F = \text{The set of all letters in the word BETTER}$
 $= \{B, E, T, R\}$

Question

Write the following sets in the set-builder form:

- (i) $\{3, 6, 9, 12\}$ (Home work)
- (ii) $\{2, 4, 8, 16, 32\}$
- (iii) $\{5, 25, 125, 625\}$
- (iv) $\{2, 4, 6, \dots\}$
- (v) $\{1, 4, 9, \dots, 100\}$

Question

$$-0.5 < x < 4.5$$

List all the elements of the following sets:

(i) $A = \{x : x \text{ is an odd natural number}\} = \{1, 3, 5, 7, 9, \dots\}$

(ii) $B = \{x : x \text{ is an integer, } -\frac{1}{2} < x < \frac{9}{2}\} = \{0, 1, 2, 3, 4\}$

(iii) $C = \{x : x \text{ is an integer, } x^2 \leq 2\} = \{-1, 0, 1\}$

(iv) $D = \{x : x \text{ is a letter in the word "LOYAL"}\} = \{L, O, Y, A\}$

(v) $E = \{x : x \text{ is a month of a year not having 31 days}\} = \{$

(vi) $F = \{x : x \text{ is a constant in the English alphabet which precedes } k\}.$

$$\{B, C, D, F, G, H, J\}$$

Question

Match each of the set on the left in the roster form with the same set on the right described in set-builder form:

- | | | |
|------------------------------------|---------------|--|
| (i) $\{1, 2, 3, 6\}$ | $\rightarrow$ | (a) $\{x : x \text{ is a prime number and a divisor of } 6\}$ |
| (ii) $\{2, 3\}$ | $\rightarrow$ | (b) $\{x : x \text{ is an odd natural number less than } 10\}$ |
| (iii) $\{M, A, T, H, E, I, C, S\}$ | $\rightarrow$ | (c) $\{x : x \text{ is natural number and divisor of } 6\}$ |
| (iv) $\{1, 3, 5, 7, 9\}$ | $\rightarrow$ | (d) $\{x : x \text{ is a letter of the word MATHEMATICS}\}.$ |

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Thank
You

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